Gaussian clock-domain obstructions for reflected-clock Brownian motion

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Reviewed editorial draft, October 29, 2026

Abstract

Let B, W be independent Brownian motions and $Z_t = B(|W_t|)$. At each positive time the actual point observable belongs to the scenery Gaussian Sobolev domain, with an explicit finite-energy gradient, but does not belong to the original clock Gaussian Sobolev domain. The bounded smooth observable $\sin Z_t$ has the same domain separation, so unbounded point growth is not the cause. The obstruction is proved in the scenery-valued Hilbert space: its clock section has squared increment norm proportional to distance, incompatible with a common locally finite-energy weak derivative. A conditional contraction of the full clock gradient would recover precisely such a derivative, so separate scalar identities cannot evade the obstruction. For a closed densely defined Hilbert operator, strong value approximation of a point outside its domain forces all derivative norms to escape to infinity, while the embedded derivatives converge in the dual of its adjoint graph with an explicit error bound. Applied to the ambient clock operator, this gives a nonrepresentable graph-dual derivative of Z_t. The conclusions concern the stated original domains and do not supply an observed-filtration stochastic integral or a physical-time Itô formula. The proofs reconstruct frozen Lean results using classical Gaussian and closed-operator methods; novelty remains unresolved.

1 The actual process and the two Gaussian domains

On a product of independent Wiener probability spaces $(\Omega_B, \mathbb{P}_B)$ and $(\Omega_W, \mathbb{P}_W)$, let B_s, W_t, $s, t \geq 0$, be standard Brownian motions starting at zero. We study

$$Z_t(\omega_B, \omega_W) = B_{|W_t(\omega_W)|}(\omega_B). \tag{1}$$

This is the simple Brownian-time Brownian motion of Allouba–Zheng [1]. The name "reflected-clock" specifies that the same scenery B is used at every visit to a given nonnegative clock level. It does not mean an excursion-by-excursion replacement of scenery or a two-sided scenery evaluated at the signed clock.

Set

$$E = L^2(\mathbb{P}_B \otimes \mathbb{P}_W), \qquad H_B = H_W = L^2(\mathbb{R}_+; ds).$$

The isonormal coordinates of B and W identify their Cameron–Martin derivative directions with H_B and H_W. Write

$$D_B : D(D_B) \subset E \longrightarrow L^2(\Omega_B \times \Omega_W; H_B), \qquad D_W : D(D_W) \subset E \longrightarrow L^2(\Omega_B \times \Omega_W; H_W)$$

for the original closed partial Gaussian derivatives. The other source is held as an L^2 parameter in each case; these are not derivatives in physical time t. Section 2 specifies their closure and fiberwise domains precisely.

Theorem 1.1. *For every $T > 0$, $Z_T \in D(D_B)$ and*

$$D_B Z_T = \mathbf{1}_{[0,|W_T|]} \quad \text{in } H_B, \qquad \|D_B Z_T\|_2^2 = \|Z_T\|_2^2 = \sqrt{2T/\pi}. \tag{2}$$

In contrast, $Z_T \notin D(D_W)$. Consequently the point is outside any joint derivative whose domain is contained in $D(D_B) \cap D(D_W)$, including the original joint derivative restricted to IBM-observable L^2 classes in the mapped development. At $T = 0$ both derivatives of $Z_0 = 0$ exist and are zero.

Theorem 1.2. *For every $T > 0$, $S_T = \sin Z_T$ is a bounded actual observable in $D(D_B)$, with*

$$D_B S_T = \cos Z_T \, \mathbf{1}_{[0,|W_T|]}, \qquad \|D_B S_T\|_2^2 \le \sqrt{2T/\pi}.$$

Nevertheless $S_T \notin D(D_W)$ and it is outside the original joint intrinsic domain. Thus that domain does not contain all bounded smooth functions of actual point observations.

We will also prove the following quantitative form of the distinction between the original graph and a weak test-dual extension.

Theorem 1.3. *Let $\mathcal{T} = D(D_W^*)$ with norm $\|v\|_{\mathcal{T}}^2 = \|v\|_2^2 + \|D_W^* v\|_2^2$. For either $f = Z_T$ or $f = S_T$ there is a bounded clock graph-dual derivative*

$$d_W f(v) = \mathbb{E}[f D_W^* v], \qquad \|d_W f\|_{\mathcal{T}^*} \le \|f\|_2.$$

For $T > 0$ no $V \in L^2(\Omega_B \times \Omega_W; H_W)$ represents it as $\mathbb{E}\langle V, v\rangle_{H_W}$ for all $v \in \mathcal{T}$. Whenever $u_n \in D(D_W)$ and $u_n \to f$ in E,

$$\|D_W u_n\|_2 \longrightarrow \infty, \tag{3}$$

$$\|j D_W u_n - d_W f\|_{\mathcal{T}^*} \le \|u_n - f\|_2 \longrightarrow 0, \tag{4}$$

where $jV(v) = \mathbb{E}\langle V, v\rangle_{H_W}$.

This theorem concerns the ambient original clock operator. Its tests can depend on both sources and are not identified with temporal predictable integrands or with the natural filtration of Z. The native point-domain obstruction in Theorem 1.1 is a separate consequence of domain inclusion.

2 Closed Gaussian derivatives and their fiberwise lift

Here are the Gaussian foundations used in the proof. For a real separable Hilbert space H, let $X(h)$ be an isonormal Gaussian family, with $\mathbb{E}X(h)X(k) = \langle h, k\rangle_H$. On finite polynomial cylinders define

$$D\, p(X(h_1), \dots, X(h_n)) = \sum_{i=1}^{n} (\partial_i p)(X(h_1), \dots, X(h_n)) h_i. \tag{5}$$

One may use orthonormal h_i after a finite linear change of coordinates. Polynomial cylinders are dense in Gaussian L^2. For completeness, in finite dimension if f is orthogonal to every polynomial, the entire transform $\int f(x) e^{z\cdot x}\, d\gamma_n(x)$ has all Taylor coefficients zero. Cauchy–Schwarz and Gaussian exponential moments justify its entireness and termwise differentiation. Its restriction to imaginary z is the Fourier transform of $f\gamma_n$, so Fourier uniqueness gives $f = 0$. In countably many coordinates, conditional expectations onto the first n coordinates converge in L^2, reducing density to the finite case.

Gaussian integration by parts gives, for polynomial cylinders p, a and a finite coordinate vector h,

$$\mathbb{E}[\langle Dp, h\rangle a] = \mathbb{E}[p\{X(h)a - D_h a\}]. \tag{6}$$

This proves closability of (5). Indeed if $p_n \to 0$ and $Dp_n \to V$ in their L^2 spaces, the right side tends to zero for each fixed a, h. Such scalar-polynomial vector tests are dense in $L^2(\Omega; H)$, so $V = 0$. The original Gaussian derivative is the closure of this graph. Its domain is dense by the preceding density argument. Smooth compactly supported coordinate tests also satisfy (6), by ordinary finite-dimensional integration by parts and passage through the closed graph. These standard source-space constructions are the setting of Gaussian Malliavin calculus; see [2] for the corresponding Gaussian-process conventions.

For an independent parameter probability space (S, ρ), the partial derivative on $L^2(S; L^2(\Omega))$ has the exact domain

$$\begin{aligned} D(\widetilde{D}) = \{f :&\text{there exists } V \in L^2(S; L^2(\Omega; H)) \text{ with} \\ &(f(s), V(s)) \in \operatorname{graph} D \text{ for } \rho\text{-almost every } s\}. \end{aligned} \tag{7}$$

The output V is unique, since the graph has a unique value in each fiber. This is the actual Bochner graph lift, not a definition using independently chosen scalar derivatives.

The lift is closed: from a convergent sequence of both graph coordinates, take a common subsequence converging almost everywhere in both fiber norms, and use closedness of the original graph in each fiber. It is densely defined because simple S-functions with values in $D(D)$ are dense in the Bochner L^2 space. It is also the closure of tensor cylinders. To see the nontrivial inclusion, a graph pair in (7) is a strongly measurable, square-integrable function into the closed Hilbert graph of D. Approximate it by simple graph-valued functions, and approximate each of their finitely many values by a polynomial graph pair of D. This yields tensor-cylinder approximation in both graph coordinates. No measurable selection of derivative values is required: the given pair already supplies the measurable output.

Taking $S = \Omega_W$ or Ω_B defines D_B and D_W as used above. Identifying product scalar L^2 with either Bochner order is an isometric Fubini identification; changing the order does not change the product law or the graph condition.

3 Finite scenery energy at a random clock level

Let $I_B : H_B \to L^2(\mathbb{P}_B)$ be the Wiener integral, so $B_s = I_B(\mathbf{1}_{[0,s]})$. Fix $s \geq 0$ and an orthonormal basis (e_j) of H_B. The partial sums

$$B_s^{(N)} = \sum_{j=1}^{N} \langle \mathbf{1}_{[0,s]}, e_j\rangle I_B(e_j)$$

are polynomial cylinders, with gradients equal to the orthogonal projections of $\mathbf{1}_{[0,s]}$. Wiener isometry and Parseval show convergence of both values and gradients. Thus B_s is in the original scenery domain, with derivative $\mathbf{1}_{[0,s]}$.

Now put $s = |W_T|$. The functions of the clock multiplying $I_B(e_j)$ are square integrable, because

$$|\langle \mathbf{1}_{[0,|W_T|]}, e_j\rangle|^2 \leq |W_T|.$$

They are allowed L^2 parameters in the scenery lift. The random projected gradients converge to $\mathbf{1}_{[0,|W_T|]}$ in the full product L^2 field norm, by pointwise Parseval and domination by $|W_T|$.

The same is true of the random partial-sum values by conditioning on W. Closedness proves the scenery part of Theorem 1.1. Conditioning also gives

$$\mathbb{E}Z_T^2 = \mathbb{E}|W_T| = \sqrt{2T/\pi};$$

the last equality is the elementary absolute first moment of a centered Gaussian of variance T. This establishes every identity in (2).

4 A Hilbert-valued section with Brownian metric

Set $\mathcal{H} = L^2(\mathbb{P}_B)$ and, for $c > 0$, define

$$F_c(x) = B_{c|x|} \in \mathcal{H}, \qquad x \in \mathbb{R}. \tag{8}$$

It is a continuous Hilbert-valued curve. For nonnegative x, y, Wiener isometry gives the exact metric

$$\|F_c(x) - F_c(y)\|_{\mathcal{H}}^2 = c|x - y|. \tag{9}$$

In particular $F_c \in L^2(\gamma; \mathcal{H})$ for standard one-dimensional Gaussian measure γ, since $\|F_c(x)\|_{\mathcal{H}}^2 = c|x|$.

Lemma 4.1. *If a continuous Banach-valued curve F on $[a, b]$, $a < b$, satisfies $\|F(x) - F(a)\|^2 \geq c(x - a)$ for $c > 0$, then it has no representation*

$$F(x) = v + \int_a^x g(s)\, ds$$

almost everywhere on $[a, b]$ with $g \in L^2([a, b])$ and a constant vector v.

Proof. The primitive is continuous, so almost-everywhere equality with the continuous F implies equality everywhere. At a this gives $v = F(a)$. Cauchy–Schwarz for the Bochner integral gives, for $x > a$,

$$c(x - a) \leq \|F(x) - F(a)\|^2 \leq (x - a) \int_a^x \|g(s)\|^2\, ds.$$

After division by $x - a$, the right side tends to zero as $x \downarrow a$, a contradiction. □

Gaussian weighting does not remove this local obstruction: on a compact interval its density is bounded above and bounded away from zero, so Gaussian L^2 derivative energy implies Lebesgue L^2 energy there.

Lemma 4.2. *Let $\mathcal{H}$ be a real separable Hilbert space and $F, g \in L^2(\gamma; \mathcal{H})$. Suppose every scalar pair $\langle e, F\rangle, \langle e, g\rangle$, $e \in \mathcal{H}$, satisfies the closed one-dimensional Gaussian derivative identity. Then on every finite interval $[a, b]$,*

$$F(x) = v + \int_a^x g(s)\, ds$$

almost everywhere, for some $v \in \mathcal{H}$.

Proof. The scalar graph identity implies

$$\int \langle e, F(x)\rangle \{xp(x) - p'(x)\}\, d\gamma(x) = \int \langle e, g(x)\rangle p(x)\, d\gamma(x)$$

for compactly supported smooth p. If φ is a compactly supported test function in (a,b), take $p=\varphi/\rho$, where ρ is the Gaussian density. Since $\rho'=-x\rho$, $xp-p'=-\varphi'/\rho$. Thus $\langle e,g\rangle$ is the ordinary weak derivative of $\langle e,F\rangle$ on (a,b). Both are locally L^2, so the one-dimensional weak fundamental theorem gives $\langle e,F(x)\rangle = c_e + \langle e, \int_a^x g(s)\,ds\rangle$ almost everywhere. This elementary theorem follows, for example, by subtracting the displayed primitive and observing that a locally integrable distribution with zero derivative is constant; convolution with compact smooth kernels proves that assertion directly.

Choose a countable dense sequence (e_j) in $\mathcal{H}$. Intersect their full-measure equality sets and choose one point x_0 in the intersection. The vector $v=F(x_0)-\int_a^{x_0} g(s)\,ds$ realizes every constant c_{e_j}. At any other point in the intersection, the inner products with all e_j of $F(x)-v-\int_a^x g$ vanish. Density makes the vector itself zero, proving the statement. □

Combining (9) on $[0,1]$ with the two lemmas shows that F_c has no single $g\in L^2(\gamma;\mathcal{H})$ realizing all scalar closed Gaussian derivative pairs. This is a common-field obstruction. It does not assert that every individual scalar weak derivative fails to exist.

5 Recovering a common field from the full clock graph

We now connect the section obstruction to the actual infinite-dimensional clock operator. Let

$$h_T = T^{-1/2}\mathbf{1}_{[0,T]} \in H_W, \qquad G = W(h_T) = W_T/\sqrt{T}.$$

Then $\|h_T\|=1$ and G is standard Gaussian. Complete h_T to an orthonormal basis of H_W. The clock Gaussian space decomposes into the coordinate G and an independent residual Gaussian family. Under this decomposition,

$$Z_T = F_{\sqrt{T}}(G) \tag{10}$$

as an actual product L^2 class, with F valued in the scenery Hilbert space.

Suppose, for contradiction, that $Z_T\in D(D_W)$, with full clock gradient V. Contract V with h_T and average only over the residual clock coordinates:

$$g(x)(\omega_B) = \mathbb{E}_{\text{residual}}\langle V(\omega_B,x,\cdot),h_T\rangle_{H_W}. \tag{11}$$

Hilbert-valued conditional integration and Jensen give

$$g\in L^2(\gamma;\mathcal{H}), \qquad \|g\|_{L^2(\gamma;\mathcal{H})}\le \|V\|_2.$$

This is one common scenery-valued field, rather than separately chosen scalar derivatives.

For $e\in\mathcal{H}$ and $p\in C_c^\infty(\mathbb{R})$, the clock test field $e(\omega_B)p(G)h_T$ has divergence

$$e(\omega_B)\{Gp(G)-p'(G)\}.$$

The formula follows on tensor polynomial cylinders from Gaussian integration by parts. Its test field and displayed divergence are in the relevant L^2 spaces; passage through the closed clock graph extends the identity to Z_T, V. Using (10) and conditional integration in (11), it becomes

$$\int \langle e,F_{\sqrt{T}}(x)\rangle\{xp(x)-p'(x)\}\,d\gamma(x) = \int \langle e,g(x)\rangle p(x)\,d\gamma(x).$$

The proof of Lemma 4.2 therefore supplies a common vector primitive for $F_{\sqrt{T}}$ on $[0,1]$. Lemma 4.1 excludes it by (9). This proves $Z_T\notin D(D_W)$.

The original joint intrinsic derivative in the mapped library is defined on the intersection of the scenery and clock domains, restricted to the closed subspace of actual IBM-observable L^2 classes. The point Z_T is observable by definition but fails the clock domain test. Thus the joint-domain conclusion requires no assertion about a new derivative domain or a new filtration. This completes Theorem 1.1.

6 A bounded smooth observable with the same obstruction

The preceding argument is not restricted to unbounded point values. We prove Theorem 1.2 using an exact first-chaos projection. Let Π be the orthogonal projection in $\mathcal{H} = L^2(\mathbb{P}_B)$ onto the closed first Gaussian chaos $I_B(H_B)$. For $s \geq 0$,

$$\Pi \sin B_s = e^{-s/2} B_s. \tag{12}$$

Indeed, for $h \in H_B$, joint Gaussian regression and one-dimensional integration by parts give

$$\mathbb{E}[I_B(h) \sin B_s] = \langle h, \mathbf{1}_{[0,s]} \rangle \mathbb{E} \cos B_s = e^{-s/2} \langle h, \mathbf{1}_{[0,s]} \rangle.$$

For $s > 0$, regression here follows simply by splitting h into its projection onto $\mathbf{1}_{[0,s]}$ and its orthogonal complement; the latter Gaussian is independent and centered. At $s = 0$ the identity is zero. The last expression is the inner product with $e^{-s/2} B_s$, proving (12).

For $c > 0$ set $S_c(x) = \sin B_{c|x|}$ as a scenery-valued Hilbert class. It is continuous and has norm at most one. Its projected curve is

$$\Pi S_c(x) = e^{-c|x|/2} B_{c|x|}, \qquad \Pi S_c(0) = 0.$$

On $0 \leq x \leq 1$, the exact source variance implies

$$\|\Pi S_c(x) - \Pi S_c(0)\|_{\mathcal{H}}^2 = e^{-cx} cx \geq c e^{-c} x. \tag{13}$$

If S_c had a common finite-energy vector primitive, applying the bounded operator Π to its Bochner integral would produce a primitive of ΠS_c with no larger derivative energy. Lemma 4.1 excludes this by (13). The recovery argument of the preceding section, with F_c replaced by S_c, would recover just such a primitive from a full clock gradient of $\sin Z_T$. Thus $\sin Z_T \notin D(D_W)$ at every positive time, and domain inclusion gives the joint intrinsic exclusion.

For the positive scenery statement, at each deterministic s the Taylor polynomials of $\sin B_s$ and their derivatives converge in Gaussian L^2, dominated by a constant times $e^{|B_s|}$. Their gradients converge to $\cos B_s \, \mathbf{1}_{[0,s]}$. Closedness gives that actual scenery graph pair. At the random level $s = |W_T|$ its field norm squared is bounded by s, and its value is bounded by one. Measurability follows from continuity of B and the interval indicator map. The exact fiber domain (7) therefore gives the scenery graph of S_T and its norm bound in Theorem 1.2. This is not an attempted clock chain rule applied to an input outside the clock domain.

7 Graph-dual derivatives and escape of the original field norm

We give the closed-operator argument separately to make its topology and representability exact. Let $T : D(T) \subset E_0 \to F_0$ be a closed densely defined operator between real Hilbert spaces. Put

$$\mathcal{G} = D(T^*), \qquad \|v\|_{\mathcal{G}}^2 = \|v\|_{F_0}^2 + \|T^* v\|_{E_0}^2.$$

The adjoint is closed, so this is a Hilbert space. Define bounded maps

$$df(v) = \langle f, T^*v\rangle_{E_0}, \qquad jV(v) = \langle V, v\rangle_{F_0}. \tag{14}$$

Cauchy–Schwarz gives $\|df\| \le \|f\|$ and $\|jV\| \le \|V\|$. For $u \in D(T)$,

$$jTu = du. \tag{15}$$

Proposition 7.1. *The map j is injective, and $df = jV$ for some $V \in F_0$ if and only if $f \in D(T)$, in which case $V = Tf$. If $f \notin D(T)$, then $df \neq 0$.*

Proof. Write $\Gamma = \operatorname{graph} T$ in $E_0 \oplus F_0$. By the definition of the adjoint and density of $D(T)$,

$$\Gamma^\perp = \{(-T^*v, v) : v \in D(T^*)\}.$$

The identity $df = jV$ says exactly that (f, V) is orthogonal to this orthogonal complement. Since Γ is closed, $\Gamma^{\perp\perp} = \Gamma$; thus (f, V) belongs to the original graph. The converse is (15). If $jV = 0$, the same orthogonality argument puts $(0, V)$ in Γ, so $V = T0 = 0$. This also proves that $D(T^*)$ is dense in F_0. Finally $df = 0 = j0$ would put f in $D(T)$, proving the last assertion. □

For any $u \in D(T)$ and $f \in E_0$, the graph-dual error bound is immediate:

$$\|jTu - df\|_{\mathcal{G}^*} = \|d(u - f)\|_{\mathcal{G}^*} \le \|u - f\|_{E_0}. \tag{16}$$

Density of $D(T)$ supplies such approximating values for every f. This convergence is in the operator norm of the graph-test dual, not in the original field space.

Proposition 7.2. *If $u_n \in D(T)$, $u_n \to f$ in E_0, and $f \notin D(T)$, then $\|Tu_n\|_{F_0} \to \infty$.*

Proof. If the norms did not tend to infinity, some subsequence would have $\|Tu_{n_j}\| \le C$ for a finite C. For $v \in D(T^*)$ take the limit in $\langle u_{n_j}, T^*v\rangle = \langle Tu_{n_j}, v\rangle$:

$$|\langle f, T^*v\rangle| \le C\|v\|_{F_0}.$$

Since $D(T^*)$ is dense in F_0, this functional extends continuously to all of F_0. Riesz representation provides $V \in F_0$ with $\langle f, T^*v\rangle = \langle V, v\rangle$ for every adjoint-domain test. Proposition 7.1 would put f in $D(T)$, a contradiction. □

Apply these results to $T = D_W$, the closed densely defined ambient clock lift of Section 2, and $f = Z_T$ or S_T (with the subscript $T > 0$ now denoting time). Theorems 1.1 and 1.2 give nonmembership, and (14), (16) and Proposition 7.2 prove Theorem 1.3. The bound on the weak derivative is explicitly $\|d_W Z_T\| \le (2T/\pi)^{1/4}$, while $\|d_W S_T\| \le 1$. Every original-domain value approximation has divergent clock energy, although its embedded derivatives converge in a weaker, precisely defined norm.

8 Scope, antecedents and formal correspondence

The Brownian-time model is classical. Allouba and Zheng define the simple reflected-clock process and distinguish it from excursion-based versions [1]. The Gaussian derivatives here act on its independent Gaussian sources; Z itself is not asserted to be a Gaussian process. Standard Gaussian derivative/divergence conventions are discussed in [2]. The recent Brownian-time change-of-measure work [3] concerns a different question; no conclusion about all possible Brownian-time calculi or changes of law follows from the clock-domain obstruction. These comparisons do not settle novelty of the present domain-and-graph-dual formulation.

The frozen verified baseline is

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The theorem map links the exact folded-section metric, first-chaos projection and bounded sine exclusion, finite-energy primitive obstruction, weak-vector primitive, actual Bochner clock contraction and observation converse, unmollified scenery derivative, actual point-domain exclusion, and generic graph-dual representability/error/escape theorems. The ambient clock application combines these exact ingredients; it is distinguished from the library's separate native adjoint-graph-dual time-average results.

The formal Gaussian realization can contain additional independent scenery coordinates, including a negative-time branch that the reflected clock never visits. The half-line presentation above retains the same actual process and partial domains on its observables. Positive graph pairs tensor with constants in unused coordinates; conversely conditional integration over those coordinates recovers the same clock test identities by contraction. Thus no unused source is substituted for a clock derivative.

No positive-time original clock gradient of the point is obtained by regularizing its values alone. In particular strong convergence of a chosen regularized clock-gradient sequence would contradict closedness; the full norm-escape conclusion applies when the approximations are actual original graph pairs. Separate scalar tests without one common finite-energy field do not meet the recovered-field hypothesis. Neither source-variable divergence nor an adjoint-graph test functional is identified here as an observed-past physical-time stochastic integrand. The paper proves no physical-time Itô or Clark–Ocone completion, and it does not address changing-past integrated point-regularizer convergence.

References

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